

Data Visualization Platform and Tourism Resilience in the Post COVID-19 Pandemic in Chiang Mai, Thailand

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Abstract— Chiang Mai has a rich heritage and the strengths for tourism sector. Enhancing regarding tourism resilience in the post-pandemic phase should be immediately addressed. This research focus on Wellness Tourism in Chiang Mai Province and provides data visualization platform to assist stakeholders and tourist in gaining confidence after the COVID-19 Crisis. In this paper, we focused “Charming Chiang Mai” project, which is an approach for enhancing tourist resilience in the post-epidemic period through creative lanna tourism. We also provide data visualization to facilitate the Chiang Mai tourist police, the Chiang Mai Public Health Office, and the Chiang Mai tourism business association in term decision making and planning. Chiang Mai province's resilience will be enhanced through the development of a digital platform and linked industrial chains. The collaborative function of city, wellness desirability, social capital, government strategies, and innovation aids in the resilience of tourist infrastructure. This study aims to contribute to rethinking whether previous tourism ideas (creative tourism) have an impact on current tourism destinations. Data visualization can enable in the resolution of current issues and the resilience of the tourist sector.

Keywords— Data visualization, digital platform, tourism resilience, post COVID-19

I. INTRODUCTION

The new corona virus (COVID-19) outbreak, which started in China in January 2020, has made significant impact on the world economy especially in the tourism sector. Tourism is a both social and economic activity that drives development forward. This outbreak already has an impact on the travelling mechanisms, which include entrepreneurs and tourists. According to the third wave of the COVID-19 pandemic in Thailand (April 2021), a large number of new infected individuals were detected around the country. The government has provided each province with measures to evaluate the risks and suitability of implementing various control measures. Due to people's uncertainty about their safety, both situations and other measures enhance the quality of life and livelihood of the people as a whole, as well as entrepreneurs and enterprises such as hotels, restaurants, and alternative tourism activities. Therefore, people stayed at home, avoided travel, and reduced their purchasing. followed by an extensive chain of subsequent effects. Due to the rapid increase of the crisis, Chiang Mai is an important tourism region in Thailand with a large number of tourism resources, including major tourist sites, operators, and tourism professionals, as well as a variety of travel service providers. This COVID-19 virus is rapidly spreading because fewer people are traveling and spending money on tourism. Therefore, it is appropriate to use the pandemic issue as a

chance to transform the tourism industry from a special emphasis on recreation to one that emphasizes physical and mental health restoration. In addition, Chiang Mai is a well-known tourist destination with comprehensive facilities, abundant tourism resources, and contemporary health care facilities. The tourism industry business sector in Chiang Mai has been disturbed since the first quarter of 2020 due to different factors of the COVID-19 crisis, as shown in Figure 1 that compares the statistics of tourists in Chiang Mai from 2019 to 2021.



Fig. 1: Statistics of tourist in Chiang Mai province since 2020-2021.

The public and private sectors of Chiang Mai are subject to formal a sandbox for tourists with two different systems: one for domestic tourists and one for foreign tourists along a specified path that limits independent travel. In this paper, we called “Charming Chiang Mai”. There is a security system based on Standard Operation Procedures (SOP) and the objective is to open tourism in all 25 districts of the province by October 2021.

We also adopted the Diamond Model of National Advantage to assist in the recovery of tourism following the COVID outbreak. This model consists of four interdependent operating environment factors that influence capacity building included: factor endowment, demand conditions, related and supporting industries and firm Strategy, structure and rivalry, as illustrated by Figure 2.

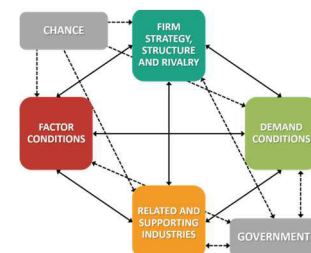


Fig. 2: Diamond Model to support tourism resilience (Porter, M.E. (1990)).

In addition to the four primary factors listed above, businesses must also consider chance and the government's role, because the government's role is essential to the country's competitive advantage. To offer a competitive advantage, the government can assess the conditions of the involved value chain and the opportunities presented by the country's environment.

The research project has developed a data visualization platform to facilitate decision-making in order to formulate measures to increase the confidence of Thai and international tourists. Therefore, numerous government and private tourism agencies in Chiang Mai, led by the Tourism Authority of Thailand and the Ministry of Public Health, engage to change the issue into an opportunity for a new tourism that would improve the business sector's resilience or the sustainable way.

II. RELATED WORK

Data Visualization played a significant role in big data because it allows the real - time representation of results and their sequences with the actual information. Data visualization facilitates data interpretation by presenting reports with objectives.

A. Visual Analytics

Visual Analytics is a technique for data analysis that combines machine learning, statistical, and visualizations. Applications of visual dashboards include, in addition to automated analysis, humans engagement providing functionality and evaluation during experiments in order to create key information for decision making [1,2]. Several articles on visual analytics have been undertaken over the years. The majority of applications relate with the visual analytics method as depicted by Keim et al[1] .[3] introduced a system for the automatic decision-making process of selecting the best pattern of visualization. This technique provided answer to the research issues: Visualization selection, which efficiently computes the top-k visualizations for an available data set by examining algorithms. The Visual Analytics platform [4, 5, 6] is divided into four core concepts: Data, Models, Knowledge, and Visualization.

B. Data visualization domain

Using an application platform enables the development of traditional contexts into innovative, intelligent, and autonomously sectors. Recently, many typical sectors, including healthcare, education, industrial, logistics, urban and sustaining management, and farming, had adopted IoT or Web-based applications. In Healthcare, a Data Visualization platform is used to accumulate actual medical data in order to provide meaningful clinical information and reduce the constraints of standard medical therapy [7]. Healthcare data collected by Internet Of Things (IOT is archived and processed on the cloud for the benefit of various parties). Thus, accessible healthcare applications can help control a patient's health status. In hospital, data analysis and visualization usually display different trends in data, providing significant information for improved decision-making and cost-effectiveness. Numerous healthcare applications link browser using GIS to enable the real-time visualization. In Smart Farming, visualization technologies allowed long-term surveillance of parameters on the farm by evaluating data collected through IoT smart sensors. Control and monitoring systems are required for growth and yield

control, plant abnormal identification, and the improvement of disease and insect prevention activities [8].

C. Tourism resilience

Some authors [9,10] indicated that businesses might be more resilient when their understanding, limitation available, and development-related enablers' capacities are increased. Tourism is highly influenced by social, culture, and economic factors; therefore, enhancing tourism's resilience is extremely dependent on a variety of sustainable development elements [11]. Individuals, businesses, industry (chains), communities, and networks will all be involved in resilience messages for the purpose of enhancing tourism's resilience in the context of COVID-19. First, from the mindset of the achievements of individuals to crisis and adaptive capacity in tourism, specific resilience plays a crucial role in driving productivity and sustaining business development [12–14], and traditional individuals are more likely to experience issues in the context of COVID-19, economic stagnation, and increased depression [15]. In other aspects, the mental health of a person, mainly those who have a role in the industry chain and structure, will influence the pandemic resistance of the tourism industry [16]. Second, from the standpoint of businesses and industrial chains, economic slowdowns, uncertainties, and advanced technology have heightened rivalry among businesses during the outbreak [17].

III. METHODOLOGY

This section describes the proposed data visualization platform to enhance tourism resilience..

A. System design

Our approach created dashboard system, which develop guidelines for managing the tourism sector after the Covid-19 crisis, a prototype of a data management system to support decision-making on city reopening measures, and a prototype area to promote Chiang Mai Medical & Wellness City including the development of partnership such as the Chiang Mai tourist police, the Chiang Mai Public Health Office, and the Chiang Mai tourism business association (see Fig.3.). It also increases capacity and hygienic safety requirements in order to maintain the confidence of tourists following the COVID-19 outbreak.

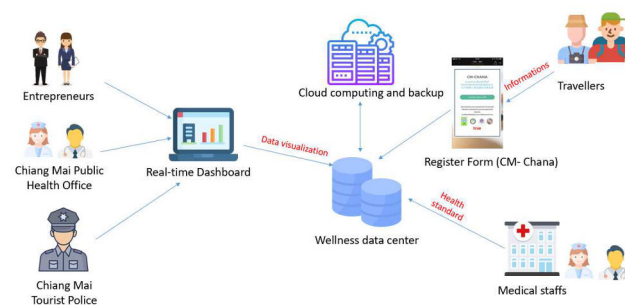


Fig. 3: Overview Data Visualization Platform in Chiang Mai.

We provided digital platform and collected data form travellers, who visited Chiang Mai province by "CM-CHANA" web-based application. It also provide data visualization to set up policy including quarantine facilities (Hospital), strengthening surveillance, high risk area, density of tourists, nationality, etc.

TABLE I. DATA VISUALIZATION FEATURES WITH SIGNIFICANT ASSOCIATES

Organization	Features /Descriptions
Chiang Mai Public Health Office	Set guidelines and policies in accordance with Thailand's Ministry of Public Health, and make decisions to regulate the COVID-19 situation in Chiang Mai province.
Chiang Mai Tourist Police	Shows the results of number of tourist as a whole or divided into groups of Thais, Asian countries, European countries, and other groups, as well as the distribution of various districts in Chiang Mai.
Medical staffs	Set up Intensive investigation, Intensive treatment and Law enforcement
Chiang Mai Tourism Business Association / Entrepreneurs	Adopt medical and wellness tourism standards to enhance capacity in Chiang Mai's medical and wellness city, also known as "Charming Chiang Mai."

B. Data Visualization to enhance policy

In this section, we represented a tool that can display information for making decision to travel, enabling travellers to visit Chiang Mai with confidence. The system is a responsible web application that is functional with all platforms, including mobile technology with IOS and Android. Big data is used to displayed all information requires. We designed a visual display of the number of cases by district's population and provided it more precise (Sub districts). It is possible to include more information on one map. Real-time visualization facilitated decision making. Dashboard displayed information from CMC-19 and CM-CHANA databases (e.g., number of cases, number of tourists, address, population, vaccine, etc.). Information will enable users at all levels of organizations in Chiang Mai to make the best decisions, both supporting individual and group decision-making, which will deliver policies and rules to facilitate the control of the COVID-19 epidemic and Chiang Mai's economic recovery.

IV. EXPERIMENTAL

We developed a system to control the spreading of COVID-19 including patient screening, surveillance, treatment, and follow-up, as well as health promotion, so that people in Chiang Mai could adjust to living in the post-epidemic period. After the relaxed situation has allowed the city accessible to both Thai and foreign tourists, Chiang Mai must plan and make strategic decisions for the return of the new pandemic while promoting tourism. Moreover, this platform can be achieved to aid in the management of information and plans for the province, whether it be the monthly tracking of outbreak data for districts (See Fig.4) or the demonstrated of tourist information for tourist police organizations categorized by nationality and by districts (See Fig.5-6). In the end, the provincial government devised regulations in accordance with the time and planned policies to efficiently serve tourists in Chiang Mai.

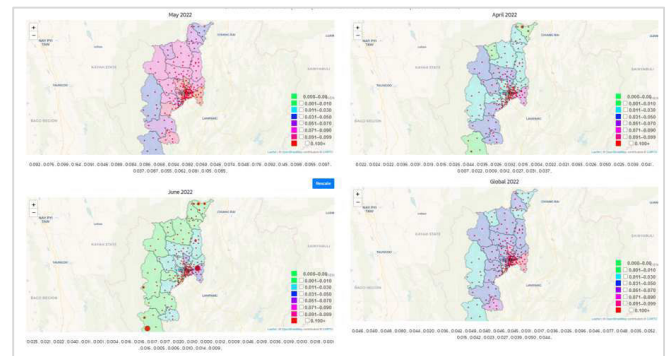


Fig. 4: Geolocalization by zone surveillance.

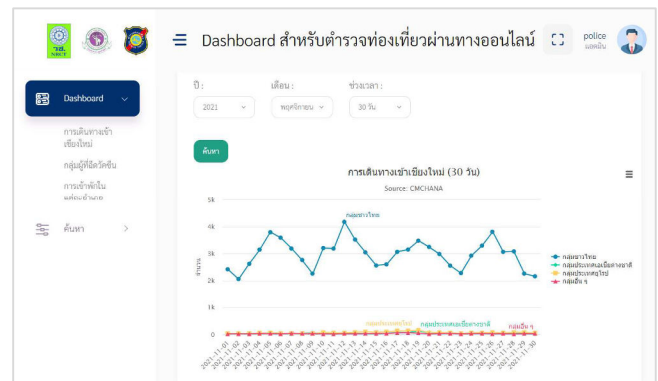


Fig. 5: Tourist information dashboard divided by nationality for tourists.

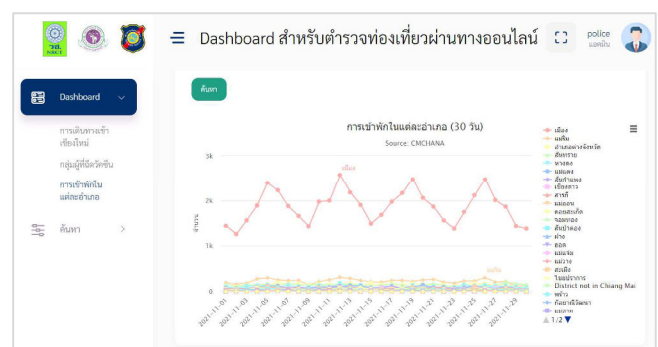


Fig. 6: Tourist information dashboard divided by districts.

We utilized the enhanced surveillance for early case detection to strengthen surveillance among high-risk groups. All cases of covid-19 were recorded in our system. This system represented Geolocalization with Time-series data Visualization (see Fig.7). Each period, the public health office can use information to determine policies that respect to Thailand's guidelines and are fit for the Chiang Mai province(see Fig.8).



Fig. 7: Risk area and time series integration.

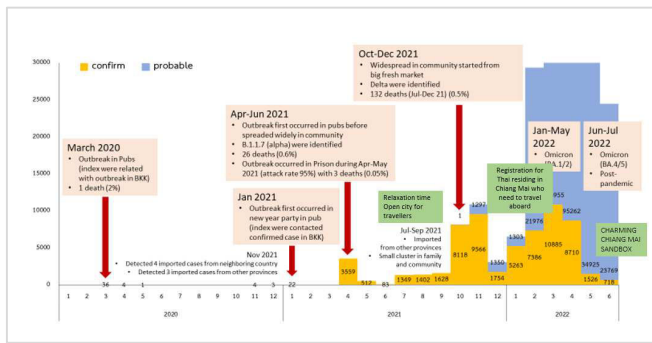


Fig. 8: Chiang Mai COVID-19 Situation, 2020-2022, Chiang Mai Policy, and tourism resilience.

V. CONCLUSION

Our method can help to maximize effectiveness policies for tourism resilience in the post COVID-19 pandemic in Chiang Mai. This research found that the implementation and expansion of the visualization platform and cultural chain could assist the provincial of Chiang Mai in strengthening its tourist resilience. Government organizations can also play a role in tourist development by selecting the appropriate data required to inform decisions. Regarding the high burden of the pandemic on tourism and factors such as cost and cost-effectiveness, the selection of tourism resilience should prioritize partnerships or stakeholder as well as a technology platforms. Future economic development can be facilitated by integrating effective policies.

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